$$
\begin{aligned}
&C_{H,1} \to \frac{1}{16\pi^2}\Big(-\frac{3}{128}\frac{1}{v_s^4} s_4{}^2 \left(g_1^2+g_2^2\right)^2 + \frac{1}{4} g_2^4 LF_{3,0}[m_2] + \frac{1}{4} g_2^4 LF_{4,-1}[m_2] - \frac{4}{15} g_2^4 LF_{5,-2}[m_2] + \\
&\frac{1}{216}\sum_p g_1^4 \left(4-9 c_2\right)^2 LF_{3,0}\left[m_{\tilde{q}}^p\right] - \frac{1}{144}\sum_p g_1^4 \left(5+6 c_2\right)^2 LF_{4,-1}\left[m_{\tilde{q}}^p\right] + \\
&\frac{1}{135}\sum_p g_1^4 LF_{5,-2}\left[m_{\tilde{q}}^p\right] - \frac{1}{3} c_2 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pr} LF_{3,0}\left[m_{\tilde{e}}^r\right] + \frac{1}{3} c_2 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pr} LF_{4,-1}\left[m_{\tilde{e}}^r\right] + \\
&\frac{1}{72}\sum_p g_1^4 \left(4+9 c_2\right)^2 LF_{3,0}\left[m_{\tilde{e}}^p\right] - \frac{1}{48}\sum_p g_1^4 \left(5+6 c_2\right)^2 LF_{4,-1}\left[m_{\tilde{e}}^p\right] + \frac{2}{45}\sum_p g_1^4 LF_{5,-2}\left[m_{\tilde{e}}^p\right] - \\
&\frac{1}{2} c_2 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pr} LF_{3,0}\left[m_{\tilde{l}}^p\right] + \frac{1}{2} c_2 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pr} LF_{4,-1}\left[m_{\tilde{l}}^p\right] + \\
&\frac{1}{144}\left(36 c_2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pr} \left(g_1^2+g_2^2\right) + \sum_p \left(g_1^4 \left(4+9 c_2\right)^2 + 3 g_2^4 \left(4-3 c_2\right)^2\right)\right) LF_{3,0}\left[m_{\tilde{l}}^p\right] + \\
&\left(-\frac{3}{4} c_2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pr} \left(g_1^2+g_2^2\right) - \frac{1}{96}\sum_p \left(g_1^4 \left(5+6 c_2\right)^2 + 3 g_2^4 \left(5-2 c_2\right)^2\right)\right) LF_{4,-1}\left[m_{\tilde{l}}^p\right] + \\
&\frac{1}{45}\sum_p \left(g_1^4 + 3 g_2^4\right) LF_{5,-2}\left[m_{\tilde{l}}^p\right] + \left(-\frac{1}{4} c_2 \left(c_\gamma^2 \overline{y_d}^{pr} y_d^{pr} \left(g_1^2 - 3 g_2^2\right) + s_\gamma^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^2 + 3 g_2^2\right)\right) + \right. \\
&\left.\frac{1}{432}\sum_p \left(g_1^4 \left(4+9 c_2\right)^2 + 27 g_2^4 \left(4-3 c_2\right)^2\right)\right) LF_{3,0}\left[m_{\tilde{q}}^p\right] + \left(\frac{1}{4} c_2 \left(c_\gamma^2 \overline{y_d}^{pr} y_d^{pr} \left(g_1^2 - 3 g_2^2\right) + s_\gamma^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^2 + 3 g_2^2\right)\right) - \right. \\
&\left.\frac{1}{288}\sum_p \left(g_1^4 \left(5+6 c_2\right)^2 + 27 g_2^4 \left(5-2 c_2\right)^2\right)\right) LF_{4,-1}\left[m_{\tilde{q}}^p\right] + \\
&\frac{1}{135}\sum_p \left(g_1^4 + 27 g_2^4\right) LF_{5,-2}\left[m_{\tilde{q}}^p\right] + \frac{1}{54}\sum_p g_1^4 \left(4+9 c_2\right)^2 LF_{3,0}\left[m_{\tilde{u}}^p\right] + \\
&\frac{1}{36}\sum_p g_1^4 \left(5+6 c_2\right)^2 LF_{4,-1}\left[m_{\tilde{u}}^p\right] + \frac{8}{135}\sum_p g_1^4 LF_{5,-2}\left[m_{\tilde{u}}^p\right] - \\
&c_2 g_1^2 s_\gamma^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,0}\left[m_{\tilde{u}}^r\right] - c_2 g_1^2 s_\gamma^2 \overline{y_u}^{pr} y_u^{pr} LF_{4,-1}\left[m_{\tilde{u}}^r\right] + \\
&\frac{1}{288}\left(g_1^4 \left(8 + 9 c_{4\gamma}\left(1 + c_{4\gamma}\right) - 9 s_{2\gamma}^4\right) + 3 g_2^4 \left(8 + 3 c_{4\gamma}\left(3 - c_{4\gamma}\right) - 3 s_{2\gamma}^4\right) + \right. \\
&\left.18 g_1^2 g_2^2 \left(c_{4\gamma}\left(-1+c_{4\gamma}\right) - s_{2\gamma}^4\right)\right) LF_{3,0}[m_S] + \frac{1}{96}\left(g_1^4 \left(5 + 3 c_{4\gamma}\left(1+c_{4\gamma}\right) - 3 s_{2\gamma}^4\right) + \right. \\
&\left.3 g_2^4 \left(-5 - c_{4\gamma}\left(-3+c_{4\gamma}\right) - s_{2\gamma}^4\right) + 6 g_1^2 g_2^2 \left(c_{4\gamma} - \left(c_{4\gamma} - 7\right)^2 + 2 s_{2\gamma}^4\right)\right) LF_{4,-1}[m_S] + \\
&\frac{1}{45}\left(g_1^4 + 3 g_2^4\right) LF_{5,-2}[m_S] + \frac{1}{16}\left(g_1^4 + 3 g_2^4\right) LF_{3,0}[\tilde{\mu}] + \frac{1}{24}\left(g_1^4 + 3 g_2^4\right) LF_{4,-1}[\tilde{\mu}] + \\
&\frac{2}{45}\left(g_1^4 + 3 g_2^4\right) LF_{5,-2}[\tilde{\mu}] + \frac{1}{8} g_1^4 \left(2 c_\gamma^4 + 3 s_\gamma^2 c_\gamma^2 + 2 s_\gamma^4\right) LF_{2,2,-1}[m_1, \tilde{\mu}] + \\
&\frac{1}{8} m_1 g_1^4 \left(m_1 \left(c_\gamma^4 + s_\gamma^4\right) + 4 s_\gamma \tilde{\mu} c_\gamma\right) LF_{2,2,0}[m_1, \tilde{\mu}] - \frac{1}{8} g_1^4 LF_{3,2,-2}[m_1, \tilde{\mu}] + \\
&\frac{3}{8} g_1^4 \left(-m_1^2 - 8 m_1 s_\gamma \tilde{\mu} c_\gamma - 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,2,-1}[m_1, \tilde{\mu}] - g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{3,2,0}[m_1, \tilde{\mu}] + \\
&\frac{1}{4} g_1^4 LF_{3,3,-3}[m_1, \tilde{\mu}] + \frac{1}{8} g_1^4 \left(m_1^2 + 8 m_1 s_\gamma \tilde{\mu} c_\gamma + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,3,-2}[m_1, \tilde{\mu}] + \\
&g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{3,3,-1}[m_1, \tilde{\mu}] + \frac{1}{8} g_1^4 LF_{4,2,-3}[m_1, \tilde{\mu}] + \\
&\frac{1}{4} g_1^4 \left(m_1^2 + 8 m_1 s_\gamma \tilde{\mu} c_\gamma + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{4,2,-2}[m_1, \tilde{\mu}] + g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{4,2,-1}[m_1, \tilde{\mu}] - \\
&g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] + \frac{3}{8} g_2^4 \left(4 c_\gamma^4 + 4 s_\gamma^2 \left(-2 + s_\gamma^2\right) + c_\gamma^2 \left(-8 + 19 s_\gamma^2\right)\right) LF_{2,2,-1}[m_2, \tilde{\mu}] + \\
&\frac{1}{8} m_2 g_2^4 \left(m_2 \left(c_\gamma^4 - 8 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) + 4 s_\gamma \tilde{\mu} c_\gamma \left(-4 + 3 c_\gamma^2 + 3 s_\gamma^2\right)\right) LF_{2,2,0}[m_2, \tilde{\mu}] + \\
&\frac{1}{4} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] - \frac{1}{8} g_2^4 \left(c_\gamma^4 - 5 c_\gamma^2 s_\gamma^2 + s_\gamma^4 + c_\gamma^2 \left(-1 + 10 s_\gamma^2\right)\right) LF_{3,2,-2}[m_2, \tilde{\mu}] + \\
&\frac{1}{8} g_2^4 \left(-3 m_2^2 \left(5 c_\gamma^4 + 2 s_\gamma^2 c_\gamma^2 + 5 s_\gamma^4\right) - 8 m_2 s_\gamma \tilde{\mu} c_\gamma \left(-1 + 9 c_\gamma^2 + 9 s_\gamma^2\right) - 36 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \\
&LF_{3,2,-1}[m_2, \tilde{\mu}] - 3 g_2^4 m_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{3,2,0}[m_2, \tilde{\mu}] + \\
&\frac{1}{4} g_2^4 \left(c_\gamma^4 + 10 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) LF_{3,3,-3}[m_2, \tilde{\mu}] + \\
&\frac{1}{4} g_2^4 \left(m_2^2 \left(5 c_\gamma^4 + 2 s_\gamma^2 c_\gamma^2 + 5 s_\gamma^4\right) + 24 m_2 s_\gamma \tilde{\mu} c_\gamma + 12 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,3,-2}[m_2, \tilde{\mu}] + \\
&3 g_2^4 m_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{3,3,-1}[m_2, \tilde{\mu}] + \frac{1}{4} g_2^4 \left(c_\gamma^4 + 10 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) LF_{4,2,-3}[m_2, \tilde{\mu}] + \\
&\frac{1}{4} g_2^4 \left(m_2^2 \left(5 c_\gamma^4 + 2 s_\gamma^2 c_\gamma^2 + 5 s_\gamma^4\right) + 24 m_2 s_\gamma \tilde{\mu} c_\gamma + 12 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{4,2,-2}[m_2, \tilde{\mu}] + \\
&3 g_2^4 m_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{4,2,-1}[m_2, \tilde{\mu}] + \frac{3}{2} c_\gamma^4 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{2,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \\
&\frac{3}{2} c_\gamma^4 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{3,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] - \\
&\frac{1}{6} g_1^2 \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}\right) LF_{2,2,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] - \\
&\frac{1}{2} c_2 g_1^2 \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}\right) LF_{3,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \\
&\frac{1}{12} g_1^2 \left(-1 + 3 c_2\right) \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}\right) LF_{3,2,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \\
&\frac{1}{2} c_2 g_1^2 \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}\right) LF_{4,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \\
&\frac{1}{2} c_\gamma^4 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{2,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t\right] - \frac{1}{2} c_\gamma^4 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{3,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t\right] + \\
&\frac{1}{2} g_1^2 \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}\right) LF_{2,2,0}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] - \\
&\frac{1}{2} c_2 g_1^2 \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}\right) LF_{3,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
&\frac{1}{4} g_1^2 \left(-1 + 3 c_2\right) \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}\right) LF_{3,2,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
&\frac{1}{2} c_2 g_1^2 \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}\right) LF_{4,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
&\frac{1}{6} \left(3 + c_2\right) \left(g_1^2 + g_2^2\right) \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}\right) LF_{4,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] + \\
&\frac{1}{8} \left(g_1^2 + 3 g_2^2\right) \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}\right) LF_{3,1,0}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] - \\
&\frac{1}{8} \left(g_1^2 \left(-5 + c_2\right) - 3 g_2^2 \left(3 + c_2\right)\right) \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}\right) LF_{4,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] + \\
&\frac{3}{2} s_\gamma^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_u}^{st} y_d^{pt} y_u^{sr} LF_{2,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] - \frac{3}{2} s_\gamma^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_u}^{st} y_d^{pt} y_u^{sr} LF_{3,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] - \\
&\frac{1}{12} \left(g_1^2 - 9 g_2^2\right) \left(s_\gamma \overline{a_u}^{pr} - \tilde{\mu} c_\gamma \overline{y_u}^{pr}\right) \left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}\right) LF_{2,2,0}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \\
&\frac{1}{24} \left(g_1^2 \left(1 + 3 c_2\right) + 9 g_2^2 \left(-1 + c_2\right)\right) \\
&\left(s_\gamma \overline{a_u}^{pr} - \tilde{\mu} c_\gamma \overline{y_u}^{pr}\right) \left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}\right) LF_{3,2,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \\
&s_\gamma^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{2,1,0}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] - \frac{3}{2} s_\gamma^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{3,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] - \\
&\frac{1}{8} g_1^4 \left(c_\gamma^4 - 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) LF_{3,2,-2}[\tilde{\mu}, m_1] + \\
&\frac{1}{8} g_1^4 \left(m_1^2 \left(c_\gamma^4 + s_\gamma^4\right) + 6 m_1 s_\gamma \tilde{\mu} c_\gamma + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,2,-1}[\tilde{\mu}, m_1] - \\
&\frac{1}{24} g_1^2 \left(g_1^2 + 3 g_2^2\right) LF_{4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{4} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \left(g_1^2 + 3 g_2^2\right) LF_{4,1,-1}[\tilde{\mu}, m_1] + \\
&\frac{1}{8} g_1^4 s_\gamma^2 c_\gamma^2 LF_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{8} g_1^4 \left(m_1^2 \left(c_\gamma^4 + s_\gamma^4\right) + 4 m_1 s_\gamma \tilde{\mu} c_\gamma + 2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{4,2,-2}[\tilde{\mu}, m_1] + \\
&\frac{1}{4} g_1^4 m_1^2 s_\gamma^2 c_\gamma^2 LF_{4,2,-1}[\tilde{\mu}, m_1] + \frac{1}{4} g_1^2 \left(g_1^2 + 3 g_2^2\right) LF_{5,1,-3}[\tilde{\mu}, m_1] + \\
&m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \left(g_1^2 + 3 g_2^2\right) LF_{5,1,-2}[\tilde{\mu}, m_1] + 2 g_2^4 LF_{2,1,0}[\tilde{\mu}, m_2] + \\
&2 m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 LF_{3,1,0}[\tilde{\mu}, m_2] + \frac{3}{8} g_2^4 \left(3 c_\gamma^4 + 8 s_\gamma^2 + 3 s_\gamma^4 + c_\gamma^2 \left(8 - 12 s_\gamma^2\right)\right) LF_{3,2,-2}[\tilde{\mu}, m_2] + \\
&\frac{1}{8} g_2^4 \left(-m_2^2 \left(c_\gamma^4 - 8 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) - 2 m_2 s_\gamma \tilde{\mu} c_\gamma \left(-8 + 3 c_\gamma^2 + 3 s_\gamma^2\right) - 12 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \\
&LF_{3,2,-1}[\tilde{\mu}, m_2] - \frac{1}{8} g_2^2 \left(5 g_1^2 + 27 g_2^2\right) LF_{4,1,-2}[\tilde{\mu}, m_2] - \\
&\frac{3}{2} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \left(g_1^2 + 7 g_2^2\right) LF_{4,1,-1}[\tilde{\mu}, m_2] + \frac{1}{4} g_2^4 \left(-2 c_\gamma^4 + s_\gamma^2 c_\gamma^2 - 2 s_\gamma^4\right) LF_{4,2,-3}[\tilde{\mu}, m_2] + \\
&\frac{1}{4} g_2^4 \left(m_2^2 \left(c_\gamma^4 - 8 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) - 12 m_2 s_\gamma \tilde{\mu} c_\gamma - 6 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{4,2,-2}[\tilde{\mu}, m_2] - \\
&\frac{3}{2} g_2^4 m_2^2 s_\gamma^2 c_\gamma^2 LF_{4,2,-1}[\tilde{\mu}, m_2] + \frac{1}{4} g_2^2 \left(g_1^2 + 3 g_2^2\right) LF_{5,1,-3}[\tilde{\mu}, m_2] + \\
&m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \left(g_1^2 + 3 g_2^2\right) LF_{5,1,-2}[\tilde{\mu}, m_2] + g_1^2 g_2^2 LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
&\frac{1}{4} g_1^2 g_2^2 \left(3 m_1 m_2 + 2 s_\gamma \tilde{\mu} c_\gamma \left(7 m_1 + 5 m_2\right) + 12 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
&2 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{2,2,1,0}[m_2, \tilde{\mu}, m_1] - \frac{1}{2} g_1^2 g_2^2 LF_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] + \\
&\frac{1}{2} g_1^2 g_2^2 \left(-m_1 m_2 - 4 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right) - 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \\
&2 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
&\frac{3}{2} c_\gamma^3 \overline{y_d}^{st} y_d^{sr} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right)\right) LF_{2,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] + \\
&\frac{3}{2} c_\gamma^3 \overline{y_d}^{st} y_d^{sr} \left(s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right)\right) LF_{3,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] + \\
&\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{2,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] - \\
&\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{3,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] + \\
&\frac{3}{4} c_\gamma^3 \overline{y_d}^{st} y_d^{sr} \left(s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{d}}^t\right] - \\
&\frac{3}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{2,2,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{d}}^t\right] + \\
&\frac{3}{2} c_\gamma^3 \overline{y_d}^{st} y_d^{sr} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right)\right) LF_{2,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \\
&\frac{3}{2} c_\gamma^3 \overline{y_d}^{st} y_d^{sr} \left(s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right)\right) LF_{3,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \\
&\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{2,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] - \\
&\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{3,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] + \\
&\frac{3}{4} c_\gamma^3 \overline{y_d}^{st} y_d^{sr} \left(s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] - \\
&\frac{3}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} LF_{2,2,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^s, m_{\tilde{d}}^r\right] + \\
&\frac{1}{2} c_\gamma^3 \overline{y_e}^{st} y_e^{sr} \left(-s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right)\right) LF_{2,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] + \\
&\frac{1}{2} c_\gamma^3 \overline{y_e}^{st} y_e^{sr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right)\right) LF_{3,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] + \\
&\frac{1}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{2,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] - \\
&\frac{1}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{3,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] + \\
&\frac{1}{4} c_\gamma^3 \overline{y_e}^{st} y_e^{sr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{e}}^t\right] - \\
&\frac{1}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{2,2,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^s, m_{\tilde{e}}^t\right] + \\
&\frac{1}{2} c_\gamma^3 \overline{y_e}^{st} y_e^{sr} \left(-s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right)\right) LF_{2,1,1,0}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
&\frac{1}{2} c_\gamma^3 \overline{y_e}^{st} y_e^{sr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right)\right) LF_{3,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
&\frac{1}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{2,1,1,0}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] - \\
&\frac{1}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{3,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] + \\
&\frac{1}{4} c_\gamma^3 \overline{y_e}^{st} y_e^{sr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] - \\
&\frac{1}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} LF_{2,2,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^s, m_{\tilde{e}}^r\right] + \\
&\frac{3}{2} s_\gamma^2 \overline{y_u}^{st} y_u^{pt} \left(-c_\gamma \overline{a_d}^{pr} + s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(-c_\gamma a_d^{sr} + s_\gamma \tilde{\mu} y_d^{sr}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r\right] + \\
&\frac{3}{2} s_\gamma c_\gamma^2 \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} \left(s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] + \\
&\frac{3}{2} s_\gamma c_\gamma^2 \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] + \\
&\frac{3}{2} \tilde{\mu} c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] + \\
&\frac{3}{2} \tilde{\mu} c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] + \\
&\frac{3}{4} s_\gamma c_\gamma^2 \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s\right] + \\
&\frac{3}{4} \tilde{\mu} c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{q}}^s\right] + \\
&\frac{3}{2} s_\gamma c_\gamma^2 \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} \left(s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \\
&\frac{3}{2} s_\gamma c_\gamma^2 \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \\
&\frac{3}{2} \tilde{\mu} c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] + \\
&\frac{3}{2} \tilde{\mu} c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] + \\
&\frac{3}{4} s_\gamma c_\gamma^2 \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] + \\
&\frac{3}{4} \tilde{\mu} c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{q}}^p\right] + \\
&\frac{3}{2} s_\gamma^3 \overline{y_u}^{st} y_u^{sr} \left(-\tilde{\mu} c_\gamma \overline{y_u}^{pr} a_u^{pt} + \overline{a_u}^{pr} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p\right] + \\
&\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{2,2,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^s\right] - \\
&\frac{1}{4} g_1^2 g_2^2 \left(c_\gamma^4 + 5 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
&\frac{1}{4} m_1 g_1^2 g_2^2 \left(m_2 \left(c_\gamma^4 - 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) - 4 s_\gamma \tilde{\mu} c_\gamma\right) LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + \\
&\frac{3}{4} g_1^2 g_2^2 \left(c_\gamma^4 + s_\gamma^4\right) LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + \\
&\frac{1}{4} g_1^2 g_2^2 \left(-m_1 m_2 \left(c_\gamma^4 - 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) + s_\gamma \tilde{\mu} c_\gamma \left(5 m_1 + m_2\right) + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] - \\
&\frac{1}{2} g_1^2 g_2^2 \left(c_\gamma^4 - s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) LF_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] + \\
&\frac{1}{4} g_1^2 g_2^2 \left(m_1 m_2 \left(c_\gamma^4 - 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) - 2 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right) - 2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \\
&\frac{1}{2} m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{4,1,1,-1}[\tilde{\mu}, m_1, m_2] - \frac{1}{2} g_1^2 g_2^2 LF_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] + \\
&\frac{1}{2} g_1^2 g_2^2 \left(-m_1 m_2 - 4 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right) - 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \\
&2 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2 LF_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] + \\
&\frac{3}{2}\left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_d}^{pr} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{st} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right) \left(-c_\gamma a_d^{sr} + s_\gamma \tilde{\mu} y_d^{sr}\right) + c_\gamma \overline{a_d}^{st}\right. \\
&\qquad \left.\left.\left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] - \\
&\frac{3}{2}\left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_d}^{pr} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{st} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right) \left(-c_\gamma a_d^{sr} + s_\gamma \tilde{\mu} y_d^{sr}\right) + c_\gamma \overline{a_d}^{st}\right. \\
&\qquad \left.\left.\left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right)\right) LF_{3,1,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] - \\
&\frac{3}{4}\left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_d}^{pr} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{st} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right) \left(-c_\gamma a_d^{sr} + s_\gamma \tilde{\mu} y_d^{sr}\right) + c_\gamma \overline{a_d}^{st}\right. \\
&\qquad \left.\left.\left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] - \\
&\frac{3}{4}\left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_d}^{pr} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{st} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right) \left(-c_\gamma a_d^{sr} + s_\gamma \tilde{\mu} y_d^{sr}\right) + \right. \\
&\qquad \left.\left.c_\gamma \overline{a_d}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_d^{pt} y_d^{sr} + c_\gamma a_d^{pt} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right)\right) \\
&\quad LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] - \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} + \right. \\
&\qquad \left.\overline{a_d}^{pr} \left(c_\gamma^2 y_d^{sr} \overline{a_d}^{st} a_d^{pt} + \overline{y_d}^{st} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right) \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right) \\
&\quad LF_{2,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} + \right. \\
&\qquad \left.\overline{a_d}^{pr} \left(c_\gamma^2 y_d^{sr} \overline{a_d}^{st} a_d^{pt} + \overline{y_d}^{st} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right) \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right) \\
&\quad LF_{3,1,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \frac{3}{4} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} + \right. \\
&\qquad \left.\overline{a_d}^{pr} \left(c_\gamma^2 y_d^{sr} \overline{a_d}^{st} a_d^{pt} + \overline{y_d}^{st} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right) \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right) \\
&\quad LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] + \frac{3}{4} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} + \right. \\
&\qquad \left.\overline{a_d}^{pr} \left(c_\gamma^2 y_d^{sr} \overline{a_d}^{st} a_d^{pt} + \overline{y_d}^{st} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right) \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^t, \right. \\
&\quad \left.m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \frac{1}{2}\left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_e}^{pr} \left(-s_\gamma \tilde{\mu} \overline{y_e}^{st} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right) \left(-c_\gamma a_e^{sr} + s_\gamma \tilde{\mu} y_e^{sr}\right) + c_\gamma \overline{a_e}^{st}\right. \\
&\qquad \left.\left.\left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \\
&\frac{1}{2}\left(-s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_e}^{pr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{st} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right) \left(-c_\gamma a_e^{sr} + s_\gamma \tilde{\mu} y_e^{sr}\right) - c_\gamma \overline{a_e}^{st}\right. \\
&\qquad \left.\left.\left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right)\right) LF_{3,1,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \\
&\frac{1}{4}\left(-s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_e}^{pr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{st} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right) \left(-c_\gamma a_e^{sr} + s_\gamma \tilde{\mu} y_e^{sr}\right) - c_\gamma \overline{a_e}^{st}\right. \\
&\qquad \left.\left.\left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] + \\
&\frac{1}{4}\left(-s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right) + \right. \\
&\quad c_\gamma \overline{a_e}^{pr} \left(s_\gamma \tilde{\mu} \overline{y_e}^{st} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right) \left(-c_\gamma a_e^{sr} + s_\gamma \tilde{\mu} y_e^{sr}\right) - \right. \\
&\qquad \left.\left.c_\gamma \overline{a_e}^{st} \left(s_\gamma^2 \tilde{\mu}^2 y_e^{pt} y_e^{sr} + c_\gamma a_e^{pt} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right)\right) \\
&\quad LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^s, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] - \frac{1}{2} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} + \right. \\
&\qquad \left.\overline{a_e}^{pr} \left(c_\gamma^2 y_e^{sr} \overline{a_e}^{st} a_e^{pt} + \overline{y_e}^{st} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right) \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right) \\
&\quad LF_{2,1,1,1,0}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \frac{1}{2} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} + \right. \\
&\qquad \left.\overline{a_e}^{pr} \left(c_\gamma^2 y_e^{sr} \overline{a_e}^{st} a_e^{pt} + \overline{y_e}^{st} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right) \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right) \\
&\quad LF_{3,1,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} + \right. \\
&\qquad \left.\overline{a_e}^{pr} \left(c_\gamma^2 y_e^{sr} \overline{a_e}^{st} a_e^{pt} + \overline{y_e}^{st} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right) \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right) \\
&\quad LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] + \frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} + \right. \\
&\qquad \left.\overline{a_e}^{pr} \left(c_\gamma^2 y_e^{sr} \overline{a_e}^{st} a_e^{pt} + \overline{y_e}^{st} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right) \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)\right) \\
&\quad LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \frac{3}{2} c_\gamma \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right) \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) \\
&\quad \left(s_\gamma \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_u}^{st} + \overline{a_d}^{pr} \left(s_\gamma \overline{a_u}^{st} - \tilde{\mu} c_\gamma \overline{y_u}^{st}\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{u}}^t\right] + \\
&\frac{3}{2} \tilde{\mu} s_\gamma^2 \overline{y_d}^{st} \overline{a_u}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right) \left(s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{u}}^r\right] + \\
&\frac{3}{4}\left(\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} \left(\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} \left(s_\gamma a_u^{sr} - 2 \tilde{\mu} c_\gamma y_u^{sr}\right)\right) + \right. \\
&\quad s_\gamma \overline{a_u}^{pr} \left(-2 \tilde{\mu} c_\gamma \overline{y_u}^{st} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right) \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) + s_\gamma \overline{a_u}^{st}\right. \\
&\qquad \left.\left.\left(\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} \left(s_\gamma a_u^{sr} - 2 \tilde{\mu} c_\gamma y_u^{sr}\right)\right)\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right]\Big)
\end{aligned}
$$